

Yik Yu Ng

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Education

Stanford University

Master of Science in Computer Science

Starting Fall 2026

Stanford, CA

McGill University

Bachelor of Science, First Class Honours in Computer Science

Aug 2023 – May 2026

Montreal, QC

Publications

Parameterized Fair Resource Allocation under Diversity Constraints

K. Huang, **Y. Y. Ng**, L. V. S. Lakshmanan, and X. Xiao

SIGKDD 2026

MADS: Ensemble LLM-based Multi-Agent Debate System for Political Bias Detection

Y. Y. Ng, B. C. M. Fung, E. Obukhova, D. Mouheb, C. Huang, and S. Huang

Submitted to DSAA 2026

GlucoseSense: Non-Invasive Blood Glucose Sensing on Mobile Devices

N. Sharma, M. Bebawy, **Y. Y. Ng**, and M. Hefeeda

MobiCom 2025

Research Experience

Research Trainee

Mila – Quebec AI Institute

May 2026 – Present

Montreal, QC

- Developing a privacy-preserving multimodal large language model (MLLM) agent system for personalized multiple-sclerosis lesion-activity prognosis from clinical narratives and MRI.
- Built a local retrieval-synthesis-critique pipeline with grounded evidence (AI co-scientist).
- Current work analyzes systematic failure subgroups using multimodal decision embeddings.
- Supervised by Tal Arbel.

Research Assistant

University of British Columbia

Jun 2025 – Jun 2026

Vancouver, BC

- Proposed a fair resource-allocation framework with a probabilistic recommendation prior, optimizing global welfare while preserving user preferences under diversity constraints.
- Developed explainable graph-learning methods for psychosis prediction from conversational text, including speech-graph construction, classification, and graph-frequency interpretation.
- Supervised by Laks V. S. Lakshmanan and Xiaokui Xiao.

Student Researcher

McGill University

Jan 2025 – Oct 2025

Montreal, QC

- Developed MADS, an ensemble multi-agent LLM debate system for detecting political bias in news articles, with paragraph-embedding-based debate routing.
- Outperformed supervised-learning baselines and larger LLMs; contributed a dataset for political-article classification.
- Supervised by Benjamin C. M. Fung and Elena Obukhova.

Research Experience (continued)

Research Assistant, NSERC USRA

May 2024 – Dec 2024

Simon Fraser University

Burnaby, BC

- Designed a mobile-device NIR data-collection protocol for non-invasive glucose concentration prediction.
- Applied deep learning to hyperspectral reconstruction and used model explanations to identify informative wavelengths for the sensing protocol.
- Supervised by Mohamed Hefeeda, Network and Multimedia Systems Lab.

Honours and Awards

Synechron Scholarship in Computer Science – Synechron Canada

Dec 2025

- \$2,500 CAD scholarship awarded by McGill's Faculty of Science for outstanding academic merit in the Honours Computer Science program.

Research Funding – DMaS Lab, McGill University

Feb 2025

- \$16,000 CAD grant supporting research at McGill's Data Mining and Security Lab.

NSERC Undergraduate Student Research Award

May 2024

- \$11,500 CAD in research support for work at Simon Fraser University.

YP Heung Foundation Post-Secondary Award

Jul 2023

- \$5,000 CAD award as one of the top six Faculty of Science students at Langara College.

References

Gladys Wong, Dr. sc. Techn. ETH Zürich

Instructor, Algorithms and Data Structures II; Department of Computer Science, Langara College

Elena Obukhova, PhD

Associate Professor of Strategy and Organization, Desautels Faculty of Management, McGill University

Independent research supervisor; faculty profile

Yvonne Leung, CFA

Head of Managed Solutions Asia, J.P. Morgan Private Bank